Section A – Answer any 2 out of 3 questions in section A

Question 1

a) Assume that there are three identical generators, each of 50 kW capacity and availability of 97% and the coincidence factor of an infinite number of households is 0.444. The peak demand of each individual household is 5 kW.

(i) If these three generators are used to supply 57 households, what is the probability to meet peak demand?

[6]

(ii) If the reliability criterion requires that the probability to meet peak demand is above 99%, how many households can be supplied by these generators?

[4]

b) Battery storage systems can provide frequency response much faster than thermal generators, therefore providing a service called Enhanced Frequency Response (EFR) in the UK. EFR reduces the need for the regular frequency response (R), following this rule

$$\left(\frac{H}{f_0} - \mathrm{EFR} \times \frac{\mathrm{T_{EFR}}}{4 \times \Delta f_{\max}}\right) \times R \geq \frac{(P_{\mathrm{L}} - \mathrm{EFR})^2 \times \mathrm{T_d}}{4 \times \Delta f_{\max}}$$

Nomenclature:

H: System level of inertia (MW-s)
R: Amount of frequency response scheduled (MW)
EFR: Amount of EFR scheduled (MW)
P_{L}: Value of generation outage we want to secure against (MW)
f_0: Nominal frequency of the power grid (Hz)
$\Delta f_{\max}$: Maximum admissible frequency deviation (Hz)
$\mathrm{T_d}$: Delivery time of frequency response (s)
$\mathrm{T_{EFR}}$: Delivery time of EFR (s)

Assumptions:
1. The system level of inertia of 150 GW-s, i.e. H = 150 GW-s,
2. The cost of frequency response: \$5/MW,
3. The largest generation unit in the UK: 1.8 GW,
4. Other numerical values: f_0 = 50 Hz, $\Delta f_{\max}$ = 0.8 Hz, $\mathrm{T_d}$ = 10 s, $\mathrm{T_{EFR}}$ = 0.5 s.

(i) If there is a bid of 150MW of EFR, what is the maximum price (£/MW) the system operator should pay for it to achieve the minimum cost?

[6]

(ii) Discuss how the maximum price (£/MW) the system operator should pay for EFR would change if (1) the system inertia level (H) decreases and (2) the amount of EFR bid increases.

[4]

Question 2

a) What are the advantages to apply per-unit system in the analysis of power systems?

[3]

b) Consider the high-voltage transmission circuit in Figure 2-1. Assume that voltages (magnitudes and phase angles) are known at both sending and receiving ends.

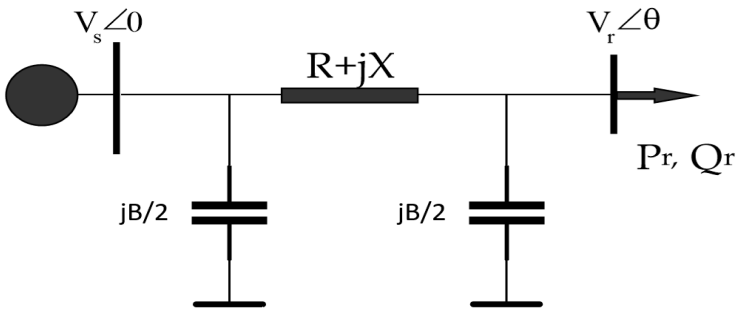


Figure 2-1 Transmission circuit with voltages specified at both sending and receiving ends

(i) Calculate the active and reactive powers in per unit at the receiving end given $V_s = 1\angle 0°$, $V_r = 1.1\angle -5°$ and $R + jX = 0.01 + j0.2$, $jB = j0.02$.

[3]

(ii) Repeat the calculation if $V_r = 0.9\angle -5°$, and comment on the influence of the difference in voltage magnitude on the power flow.

[3]

(iii) Repeat the calculation if $V_r = 1.1\angle +5°$, and comment on the influence of the difference in voltage angle on the power flow.

[3]

c) In a balanced three-phase system, assuming $V = 1\ pu$, $I = 0.5\ pu$ and $\phi = 60^o$ for each phase, calculate the three-phase active power and reactive power.

[8]

Question 3

A three-bus power network is presented in Figure 3.1 below. Data relevant for the load flow analysis on this system are given in per-unit values.

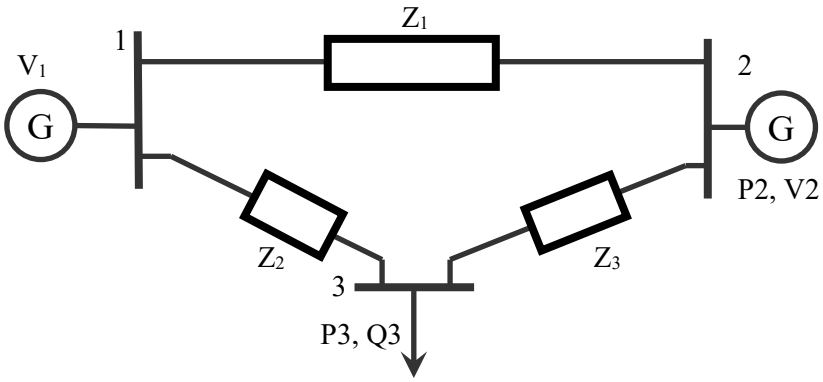


Figure 3.1 A three-bus power network

a) Explain the concept of the slack bus, PV bus and PQ bus in the load flow calculations

[3]

b) Express the bus admittance matrix for the system in terms of Z_1, Z_2, Z_3.

[5]

c) Assuming the per unit values |V1| = 1, |V2| = 1.05, P2 = 1, P3 = 1, Q3 = 1.03, $Z_1 = 0.25j$, $Z_2 = 0.25j$ and $Z_3 = 0.2j$, perform two iterations of the Gauss-Seidel load flow method.

[12]

Section B – Answer any 2 out of 3 questions in section B

Question 4

a) Consider a power system with N number of buses for which the bus impedance matrix (Z_{BUS}) and the pre-fault bus voltages ($V_1^0, V_2^0, \ldots, V_n^0$) are known. Derive the expression for the following considering a three-phase fault at bus m with a fault impedance, Z_f. The expression should be in terms of the appropriate elements of Z_{BUS}, fault impedance, Z_f and relevant pre-fault voltages.

i) Fault current at bus m

[5]

ii) Voltage at bus k (V_k^f) during the fault at bus m

[2]

b) The bus impedance matrix (Z_{BUS}) for a power system with a base MVA of 500 is given below. This includes the sub-transient reactance of the generators at the relevant buses.

$$Z_{BUS} = \begin{bmatrix} j0.14 & j0.11 & j0.13 \\ j0.11 & j0.14 & j0.13 \\ j0.13 & j0.13 & j0.18 \end{bmatrix} \text{ p.u.}$$

Calculate the magnitude of the following considering the pre-fault voltages at all the buses to be 1.0 p.u. and a fault impedance of 0.02 p.u.

i) Three-phase fault current (in p.u.) at bus 1.

[2]

ii) Voltage (in p.u.) at buses 2 and 3 during the three-phase fault at bus 1.

[4]

iii) Short circuit level (in MVA) at bus 3.

[2]

c) State the role of electrical and magnetic circuit transients in fault analysis and how these are accounted for in fault calculations.

[3]

d) Explain why the short-circuit level is declining as thermal power plants with synchronous generators are replaced by wind farms and solar photovoltaic generation.

[2]

Question 5

The sequence reactance of the generators, transformers and the line of the three-phase power system shown in Figure 5.1 are given in Table 5.1. The reactance values are in per unit (p.u.) with respect to 100 MVA base. Both generators have solidly grounded neutral.

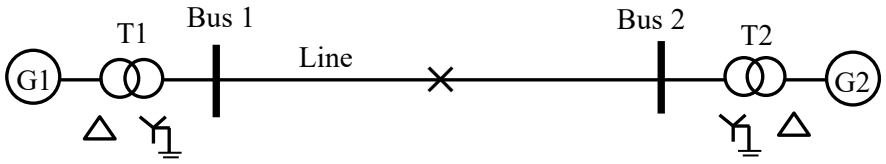


Figure 5.1: Single-line diagram of a three-phase power system

	Rated voltage	Positive seq.	Negative seq.	Zero seq.
Generators G1, G2	20 kV	$X_1'' = 0.08$ p.u.	$X_2'' = 0.08$ p.u.	$X_0'' = 0.03$ p.u.
Transformers T1, T2	20 kV/132 kV	$X_1 = 0.1$ p.u.	$X_2 = 0.1$ p.u.	$X_0 = 0.1$ p.u.
Line	132 kV	$X_1 = 0.2$ p.u.	$X_2 = 0.2$ p.u.	$X_0 = 0.55$ p.u.

Table 5.1: Parameters of the system shown in Figure 5.1

Considering the pre-fault voltages to be 1.0 p.u. and pre-fault currents to be 0, calculate the following for a line-to-ground fault on phase A at the mid-point of the line (shown by a cross in Figure 5.1) with zero fault impedance.

a) Current (in kA) flowing into the fault.

[6]

b) Phase C voltage (in kV) at the fault point.

[5]

c) How much would be the fault current (in kA) and phase C voltage (in kV) at the fault point if both windings of the transformers (T1 and T2) were star connected with a solidly grounded neutral?

[6]

d) Calculate the fault current (in kA) for a three-phase fault at the mid-point of the line.

[3]

Question 6

A synchronous generator-based power plant is connected to a large power system through a transformer and a transmission line. The parameters are provided in Table 6.1

System frequency	50 Hz
Generator synchronous reactance	0.07 p.u.
Generator inertia constant	4.0 sec
Generator terminal voltage	1.03 p.u.
Generator active power output	1.3 p.u.
Transformer leakage reactance	0.01 p.u.
Transmission line reactance	0.25 p.u.
Voltage at the large power system bus	$1.0\angle 0°$ p.u.

Table 6.1: System parameters with all per unit (p.u.) quantities expressed on 50 MVA base

a) Calculate the magnitude of the internal voltage of the generator behind its synchronous reactance. Consider this internal voltage magnitude to be held constant for the rest of the parts of this question 6.

[3]

b) Determine the critical clearing time (in milliseconds) for a temporary fault on the line. Assume zero power transfer between the generator and the large power system during the fault.

[5]

c) If the mechanical power input to the generator is suddenly increased by 20%, derive an expression for the maximum excursion point of the power angle.

[5]

d) Calculate the steady-state stability limit for each of the following cases:

i) With a series compensation level of 20%

[1]

ii) With an ideal phase angle compensation of $\pm 15°$

[1]

iii) With ideal mid-point shunt reactive compensation regulating the mid-point voltage at 1.0 p.u.

[1]

e) What would be the critical clearing time (in milliseconds) if a TCSC with a series compensation level of 20% is installed on the transmission line? Use the same assumptions from part (b).

[4]